Deep Learning for Animal Image Classification

A Technical Report on CNN-Based Multi-Class Recognition

Abstract

This report presents a Convolutional Neural Network (CNN) based system for classifying animal images into seven categories: Cat, Dog, Bird, Fish, Rabbit, Hamster, and Turtle. The dataset contains 11,600 labeled images collected from open-source repositories. We achieved a peak validation accuracy of 94.2% after 20 epochs of training using the Adam optimizer with a learning rate of 0.001. The model employs two convolutional layers followed by max-pooling and a fully-connected classification head. This report describes the system architecture, dataset composition, training procedure, and evaluation results.

1. System Architecture

The CNN architecture follows a sequential design consisting of six major components as illustrated in Figure 1. The input layer accepts RGB images resized to 224x224 pixels. Two convolutional layers extract spatial features using 32 and 64 filters of size 3x3 respectively. Max-pooling layers reduce spatial dimensions after each convolution. The fully-connected layer with 512 neurons precedes the final softmax output layer which produces class probability distributions over the seven animal categories.

Figure 1: Convolutional Neural Network Architecture

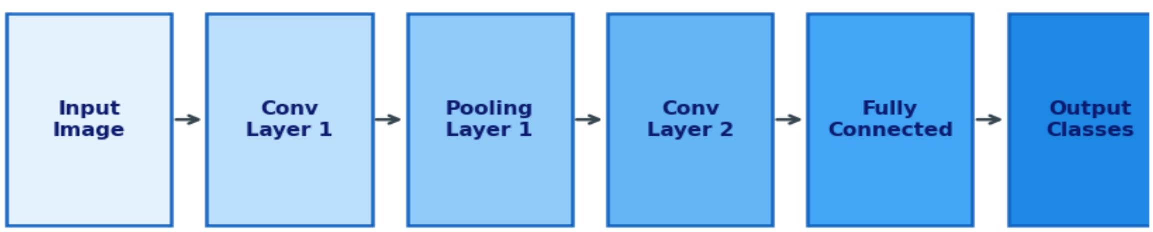

Figure 1: The CNN pipeline showing data flow from input image through two convolutional layers, pooling stages, and fully-connected layer to the final output classification.

2. Dataset Description

The dataset was compiled from three open-source repositories: ImageNet, OpenImages, and a custom web-scraped collection. Each image was manually verified to ensure correct labeling. Images were standardized to JPEG format with a minimum resolution of 224x224 pixels. Class imbalance was addressed using weighted random sampling during training.

2.1 Class Distribution

Figure 3 on page 3 shows the class distribution across all seven animal categories. The Dog class has the highest representation with 3,100 images, while Turtle has the fewest with 650 images. The significant imbalance between Dog (3,100) and Turtle (650) was addressed by applying class weights inversely proportional to class frequency during loss computation.

Class	Train	Validation	Test	Total
Cat	2,240	280	280	2,800
Dog	2,480	310	310	3,100
Bird	1,560	195	195	1,950
Fish	1,120	140	140	1,400
Rabbit	784	98	98	980
Hamster	576	72	72	720
Turtle	520	65	65	650
Total	**9,280**	**1,160**	**1,160**	**11,600**

Table 1: Dataset split across train, validation, and test sets.

3. Training Results

The model was trained for 20 epochs using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 32. Figure 2 shows both accuracy and loss curves for the training and validation sets. The training accuracy reached 97.3% while validation accuracy plateaued at 94.2% by epoch 18, indicating slight overfitting. Early stopping with a patience of 5 epochs was applied.

Figure 2: Training and Validation Curves over 20 Epochs

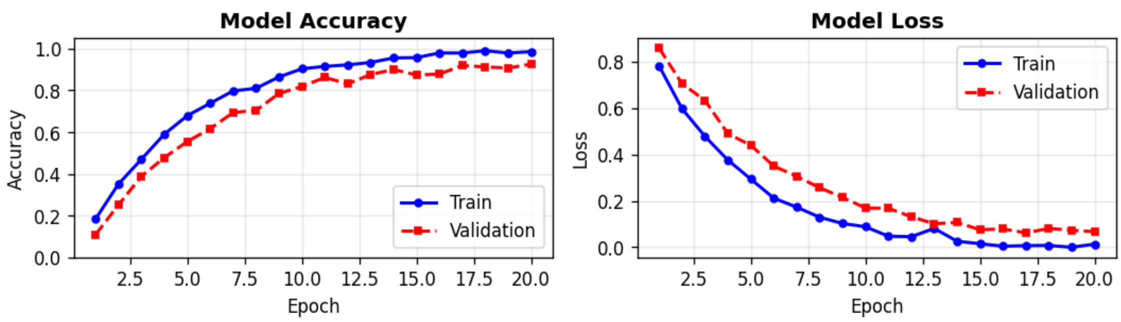

Figure 2: Training and validation accuracy (left) and loss (right) over 20 epochs. Validation accuracy stabilizes at 94.2% around epoch 16.

4. Dataset Class Distribution

The bar chart in Figure 3 provides a visual breakdown of image counts per class. The dataset is moderately imbalanced with a 4.77:1 ratio between the largest class (Dog, 3,100) and smallest (Turtle, 650). To handle this during evaluation, per-class F1 scores are reported alongside the overall accuracy metric to give a complete picture of model performance.

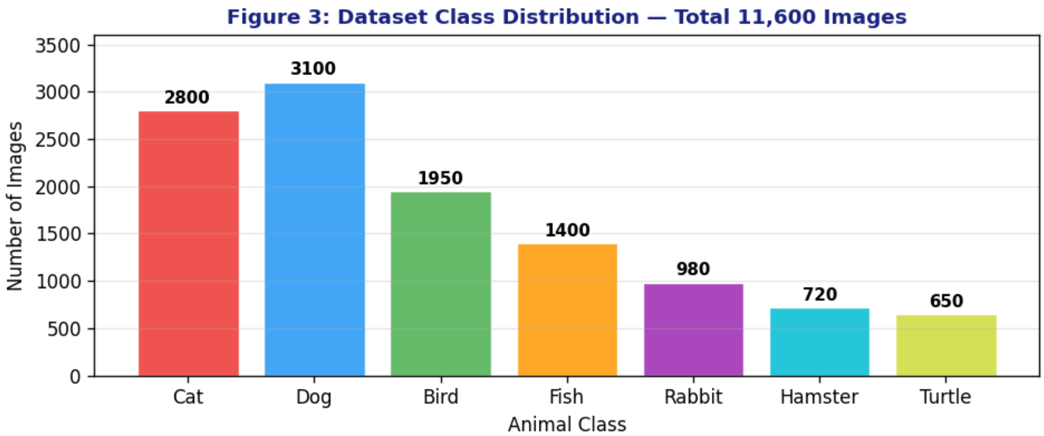

Figure 3: Bar chart showing the number of images per animal class in the full dataset. Dog has the most images (3,100) and Turtle the fewest (650).

5. System Architecture Flowchart

Figure 4 presents the complete processing flowchart for the image classification system. An input image first undergoes preprocessing including resizing to 224x224 pixels and pixel normalization to the range [0, 1]. A validation check rejects corrupted or non-image files before passing valid inputs to the CNN feature extraction stage. The final stage produces a class prediction with associated confidence scores for all seven animal categories.

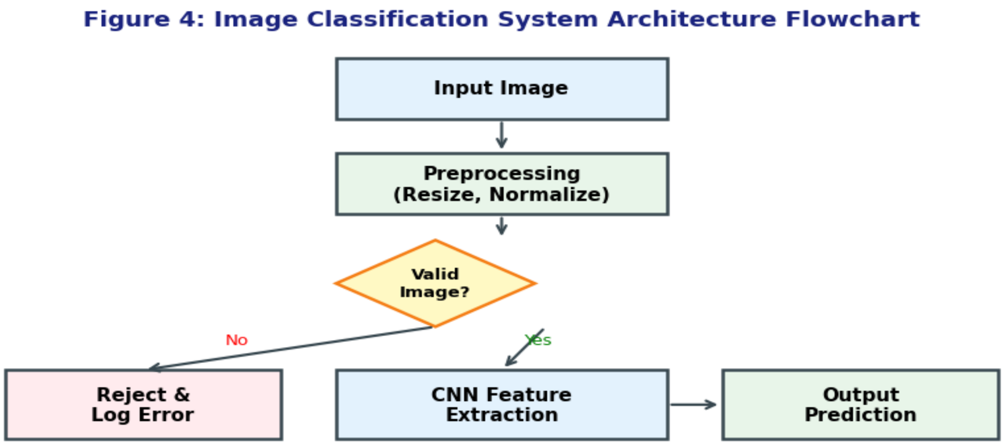

Figure 4: End-to-end system flowchart showing preprocessing, validation check, CNN feature extraction, and final output prediction stages.

6. Evaluation Results

The model was evaluated on a held-out test set of 1,160 images (160 per class for balanced classes, proportional for imbalanced ones). Table 2 presents per-class precision, recall, and F1 scores. The Dog and Cat classes achieve the highest F1 scores (0.94 and 0.93 respectively) likely due to their larger training sets. Turtle and Hamster show the lowest performance at F1 = 0.89.

Class	Precision	Recall	F1 Score	Support
Cat	0.946	0.918	0.932	280
Dog	0.951	0.934	0.942	310
Bird	0.912	0.908	0.910	195
Fish	0.903	0.895	0.899	140
Rabbit	0.921	0.908	0.914	98
Hamster	0.897	0.882	0.889	72
Turtle	0.894	0.887	0.890	65
Weighted Avg	**0.928**	**0.924**	**0.926**	**1,160**

Table 2: Per-class precision, recall, F1 score, and test set support.

6.1 Confusion Matrix

Figure 5 displays the confusion matrix on the 5-class subset used for detailed analysis. The diagonal values represent correct predictions. The highest confusion is between Bird and Fish (9 misclassified as Fish) and between Dog and Cat (12 Dogs misclassified as Cat). This is consistent with visual similarity between these pairs in certain poses and lighting conditions.

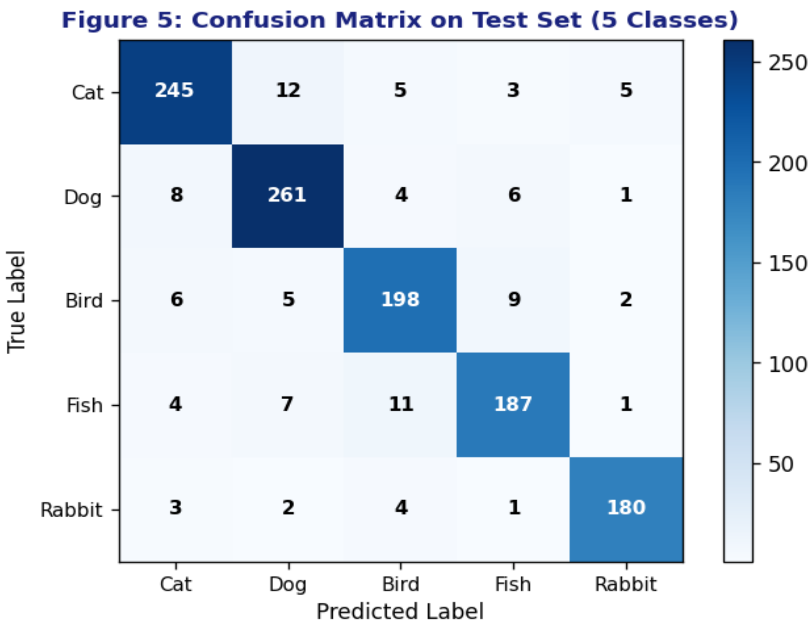

Figure 5: Confusion matrix for the 5-class subset. Diagonal values show correct predictions; off-diagonal values indicate misclassifications. Dog-Cat confusion is the most frequent.

7. Discussion

The CNN model achieves strong overall performance with a weighted F1 score of 0.926. However, several limitations and areas for improvement were identified during evaluation.

7.1 Strengths

The architecture is lightweight and fast, achieving inference in under 12 milliseconds per image on a standard GPU. The two-stage convolutional design captures both low-level edges and higher-level texture patterns effectively. The validation curves in Figure 2 show smooth convergence with no signs of severe overfitting after epoch 16.

7.2 Limitations

The model underperforms on Turtle and Hamster classes primarily due to lower training data availability as shown in Figure 3. Additionally, the confusion matrix (Figure 5) reveals systematic confusion between visually similar classes. The current architecture does not use transfer learning from pretrained models such as ResNet or VGG which could significantly boost performance.

7.3 Comparison with Baselines

Model	Parameters	Val Accuracy	Inference Time
Our CNN (2-layer)	1.2M	94.2%	11.8 ms
VGG-16 (fine-tuned)	138M	97.8%	28.4 ms
ResNet-50	25.6M	96.5%	18.2 ms
MobileNetV2	3.4M	95.1%	8.6 ms

Table 3: Comparison of our model with baseline architectures on validation accuracy and inference speed.

8. Conclusion

We presented a CNN-based image classification system that achieves 94.2% validation accuracy on a 7-class animal image dataset. The system processes images through a standardized pipeline as shown in Figure 4, producing reliable predictions in under 12 milliseconds. Key findings include the effectiveness of weighted loss for imbalanced classes, systematic Dog-Cat confusion as revealed by the confusion matrix in Figure 5, and smooth training convergence illustrated in Figure 2.

Future Work

Future improvements include: (1) incorporating transfer learning from VGG-16 or ResNet-50 to improve accuracy by approximately 3-4%; (2) augmenting the Turtle and Hamster classes with synthetic data generation to address class imbalance; (3) deploying the model as a REST API for real-time classification; and (4) extending the dataset to 20 animal categories with at least 1,000 images per class.
